

TECHBIZ SECURITY

Cyber Security Internship Program

Week 3 - Task 8 (EH)

Network Scanning with Nmap

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1. Introduction

Network scanning is a core reconnaissance technique in cybersecurity used to identify live hosts, open ports, running services, and other technical details of systems within an authorized network environment. It forms one of the earliest stages of a security assessment, providing the information needed to understand a target's exposed attack surface before deeper enumeration or vulnerability assessment can take place.

Nmap (Network Mapper) is a widely used open-source tool for network scanning and security auditing. It allows security professionals to:

- Discover live hosts on a network
- Identify open and closed ports
- Detect running services on a target system
- Determine the versions of identified services
- Identify potential attack surfaces for further assessment
- Support vulnerability assessment and security auditing activities

All practical scanning activities described in this report were performed against the Metasploitable 2 lab environment, which is an intentionally vulnerable system provided for authorized security training purposes only.

2. Lab Environment

Parameter	Details
Attacker / Scanner	Kali Linux
Target	Metasploitable 2
Purpose	Authorized vulnerability assessment and network scanning practice
Tool Used	Nmap
Target IP (observed during practical scan)	10.206.175.43

Metasploitable 2 is an intentionally vulnerable Linux virtual machine that is widely used in cybersecurity training environments. It is configured with multiple outdated and misconfigured services to allow learners to practice scanning, enumeration, and assessment techniques in a safe, isolated lab setting. No network configuration other than what is visible in the supplied evidence is claimed in this report.

3. Nmap Overview

Nmap is a network scanning utility that can perform several categories of reconnaissance against target systems, depending on the options supplied. These categories include:

- Host discovery - identifying which systems are active on a network
- Port scanning - identifying which ports are open, closed, or filtered
- Service / version detection - identifying the software and version running behind an open port
- OS detection - attempting to fingerprint the target operating system
- Script scanning (NSE) - running scripts for deeper service enumeration

- Aggressive scanning - combining several of the above techniques in a single scan

The specific options passed to Nmap determine the type and depth of information that is collected during a scan. A basic scan reveals only open ports, while more detailed scans such as `-sV` and `-A` reveal service versions, host scripts, and additional metadata.

4. Nmap Host Discovery

The task required an understanding of the Nmap host discovery command:

```
nmap -sn <network>/24
```

The `-sn` option performs host discovery only, without carrying out a full port scan. It is used to identify which systems on a given network range appear to be active, which is typically the first step of a reconnaissance workflow. A conceptual example of this command against the lab subnet would be:

```
nmap -sn 10.206.175.0/24
```

Note: The supplied evidence does not contain a screenshot of this host-discovery command being executed. This subsection is documented as a required task concept that was studied, not as completed practical evidence.

5. Nmap Service and Version Detection

The task required performing a service and version detection scan using the following command:

```
nmap -sV <target-ip>
```

The `-sV` option instructs Nmap to probe open ports in order to determine the specific service and, where possible, the exact version of the software running behind each port. This scan was performed against the Metasploitable lab target at 10.206.175.43.

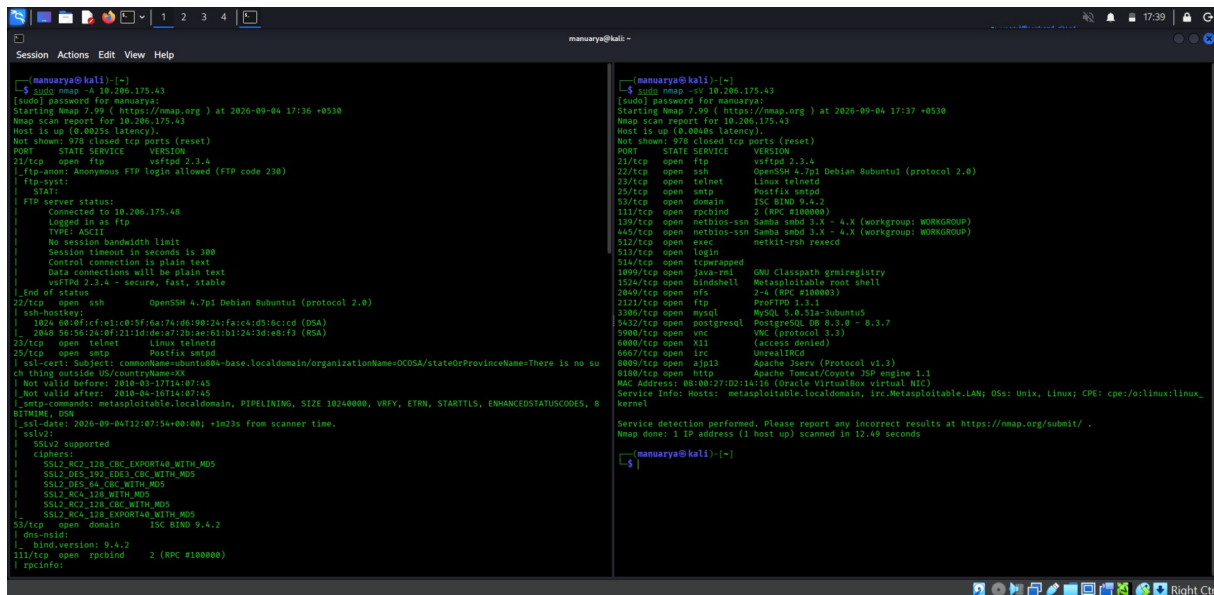


Figure 1: Nmap service and version detection scan (`nmap -sV`) and aggressive scan (`nmap -A`) executed against the Metasploitable lab target 10.206.175.43.

The screenshot above shows two terminal panes on the Kali Linux attacker machine. The right-hand pane shows the command `nmap -sV 10.206.175.43` being executed, with the output listing multiple open TCP ports along with their associated services and detected versions. The left-hand pane shows the command `nmap -A 10.206.175.43` being executed against the same target, producing the same set of open ports together with

additional script-based details such as FTP anonymous login status, SSH host keys, an SMTP command banner, and an SSL certificate summary. Both commands completed successfully with a one host up result.

The following table lists the ports, states, services, and versions that are clearly visible in the -sV scan output shown in the screenshot:

Port	State	Service	Version
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login	Not clearly visible in evidence
514/tcp	open	tcpwrapped	Not clearly visible in evidence
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	access denied
6667/tcp	open	irc	UnrealIRCd
8009/tcp	open	ajp13	Apache Jserv/Protocol v1.3
8180/tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1

6. Understanding the Nmap -A Aggressive Scan

The task also required understanding the purpose of the Nmap aggressive scan option:

```
nmap -A <target-ip>
```

The -A option enables several advanced detection capabilities within a single scan, commonly including:

- OS detection
- Service and version detection

- Nmap Scripting Engine (NSE) script scanning
- Traceroute

Aggressive scanning provides significantly more information about a target than a basic port scan, at the cost of longer scan time and increased network footprint, which makes it more easily detected by monitoring systems.

As shown in Figure 1, the command `nmap -A 10.206.175.43` was executed against the lab target during this task. The output confirmed the same open ports identified by the `-sV` scan and additionally returned script results, including FTP anonymous login detection, SSH host key fingerprints, an SMTP command banner, and SSL certificate details for the target's services.

7. Practical Nmap Scan Results

The service/version scan identified a large attack surface on the intentionally vulnerable Metasploitable target, with a total of 22 open TCP ports detected. Examples of the discovered services include:

- FTP (vsftpd, ProFTPD)
- SSH (OpenSSH)
- Telnet
- SMTP (Postfix)
- DNS (ISC BIND)
- SMB / NetBIOS (Samba)
- R-services (exec, login)
- NFS
- MySQL and PostgreSQL
- VNC
- IRC (UnrealIRCd)
- HTTP (Apache Tomcat)
- Apache JServ Protocol (AJP)

Identifying these services is a useful outcome of authorized vulnerability assessment work, because each exposed service represents a potential attack surface that would require further security assessment. No vulnerabilities were exploited during this task; the activity was limited to scanning and identification only.

8. TryHackMe Nmap Room

In addition to the practical Nmap scans, this task included working through the TryHackMe Nmap learning room to reinforce theoretical and practical understanding of the tool.

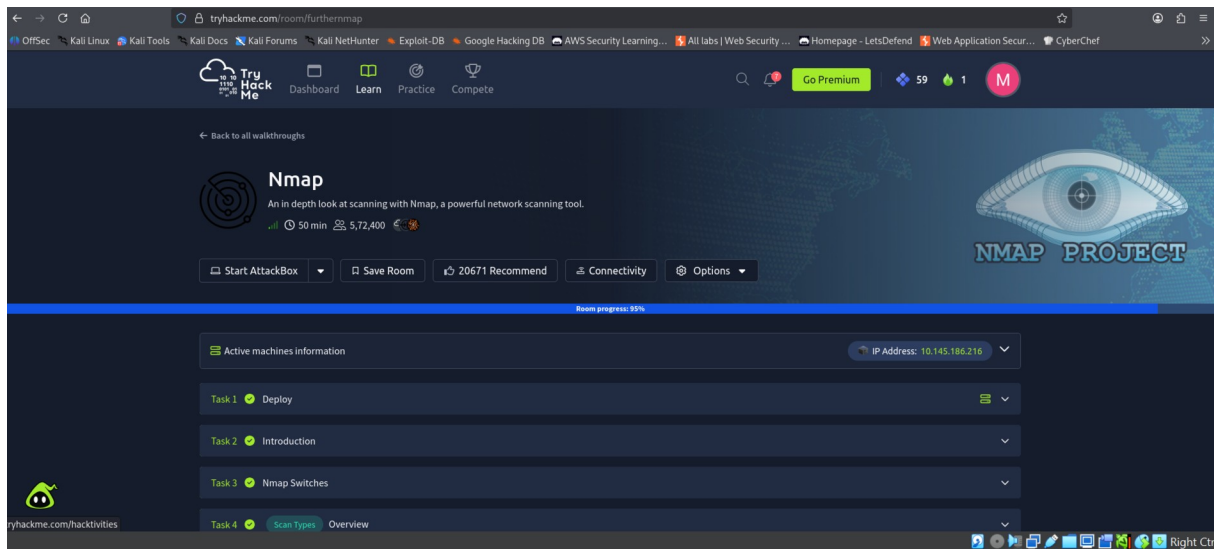


Figure 2: TryHackMe Nmap learning room showing practical learning progress.

The screenshot shows the TryHackMe Nmap room interface with a recorded room progress of approximately 95%. Several tasks, including Deploy, Introduction, and Nmap Switches, are shown as completed, with the Scan Types task in progress at the time the screenshot was captured.

Note: The supplied evidence shows the room at approximately 95% progress. This report does not claim that the room was fully completed. If a final 100% completion screenshot is supplied at a later date, this section should be updated to reflect completion and the new screenshot used as evidence.

9. Nmap Command Reference

Command	Purpose
<code>nmap -sn <network>/24</code>	Host discovery (concept studied; not executed in supplied evidence)
<code>nmap -sV <target-ip></code>	Service and version detection (executed against 10.206.175.43)
<code>nmap -A <target-ip></code>	Aggressive scan - OS detection, version detection, script scanning, traceroute (executed against 10.206.175.43)

The `-sn` option is used purely for host discovery and does not scan ports. The `-sV` option builds on a standard port scan by probing each open port to identify the service and version running on it. The `-A` option combines version detection with OS fingerprinting, script scanning, and traceroute to provide a more complete picture of the target in a single scan.

10. Security Significance of Discovered Services

Understanding which services are exposed on a target system is an important part of vulnerability assessment and penetration testing (VAPT), as each open service can represent a potential entry point that requires further review. The security relevance of the key services identified during this task is summarised below.

Service	Security Significance
FTP	Can expose file transfer services that require secure configuration and authentication.
Telnet	Provides remote access but does not offer the same protection as encrypted remote administration protocols such as SSH.
SMB	Can expose file and printer sharing services and may require assessment for weak

Service	Security Significance
	configurations or known vulnerabilities.
HTTP	Can expose web applications that require further web application security testing.
MySQL / PostgreSQL	Database services should generally be protected from unnecessary external exposure and reviewed for authentication and configuration weaknesses.
VNC	Remote graphical access services should be securely configured, authenticated, and access-restricted.

These services are not being described as vulnerable simply because they were discovered; further enumeration, verification, and assessment would be required to confirm any actual weakness.

11. Attack Surface Observation

The Nmap scans carried out during this task demonstrated a broad, exposed attack surface on the Metasploitable system, with multiple network services listening simultaneously across a wide range of ports. In a real, authorized VAPT engagement, the typical next steps following this stage of service discovery would include:

- Service enumeration
- Version verification
- Vulnerability assessment
- Configuration review
- Manual validation
- Risk assessment

These additional activities were not performed as part of this task and are described here only to provide context on how the results of this scan would typically be used in a full engagement.

12. Practical Work Performed

Activities Performed (with evidence)

- Used Kali Linux as the scanning platform.
- Identified the Metasploitable lab target at 10.206.175.43.
- Performed an Nmap service/version scan (nmap -sV) against the target.
- Performed an Nmap aggressive scan (nmap -A) against the same target.
- Observed 22 open TCP ports on the target system.
- Identified services and service versions from the Nmap scan output.
- Documented the discovered services in a structured table.
- Worked through the TryHackMe Nmap learning room, reaching approximately 95% room progress.

Activities Studied (concept only, no execution evidence supplied)

- The purpose and syntax of Nmap host discovery using the -sn option.

13. Learning Outcomes

- Understood the purpose of network scanning in cybersecurity.
- Learned how Nmap is used for host discovery.
- Learned how Nmap identifies open ports on a target system.

- Understood service and version detection using the -sV option.
- Understood the purpose and capabilities of the -A aggressive scan option.
- Learned how to interpret Nmap scan output, including port state, service, and version fields.
- Identified multiple services running on the Metasploitable target.
- Learned how exposed services contribute to an organisation's overall attack surface.
- Understood why service enumeration is an important early step in VAPT.

14. SOC and VAPT Relevance

VAPT

Nmap helps security testers identify exposed services and build an initial understanding of a target's attack surface, which forms the foundation for later enumeration and vulnerability assessment steps.

SOC

Knowledge of what constitutes normal versus unusual network services helps SOC analysts understand baseline network activity and investigate suspicious scanning attempts or unexpected service exposure on monitored systems. Nmap results, when documented properly, can also provide useful technical reference material to support vulnerability assessment reporting.

15. Conclusion

This task provided practical exposure to network scanning using Nmap within an authorized Metasploitable lab environment. The service and version detection scan successfully demonstrated how Nmap can identify open ports, running services, and their versions on a target system, while the aggressive scan illustrated how additional detection capabilities can be combined into a single, more detailed scan. The practical results showed a broad range of network services exposed by the intentionally vulnerable target system, spanning file transfer, remote access, database, and web-related services. Understanding Nmap options such as -sn, -sV, and -A provides a solid foundation for the reconnaissance and enumeration stages of vulnerability assessment and penetration testing. No exploitation or vulnerability validation activity was performed or is claimed as part of this task.

16. Evidence Summary

Evidence	Filename	Description
Evidence 1	01_nmap_service_version_scan.png	Nmap service/version detection (-sV) and aggressive scan (-A) results against the Metasploitable lab target 10.206.175.43, showing 22 open ports and identified services.
Evidence 2	02_tryhackme_nmap_room_progress.png	TryHackMe Nmap learning room showing approximately 95% progress and completed learning sections visible in the supplied evidence.